

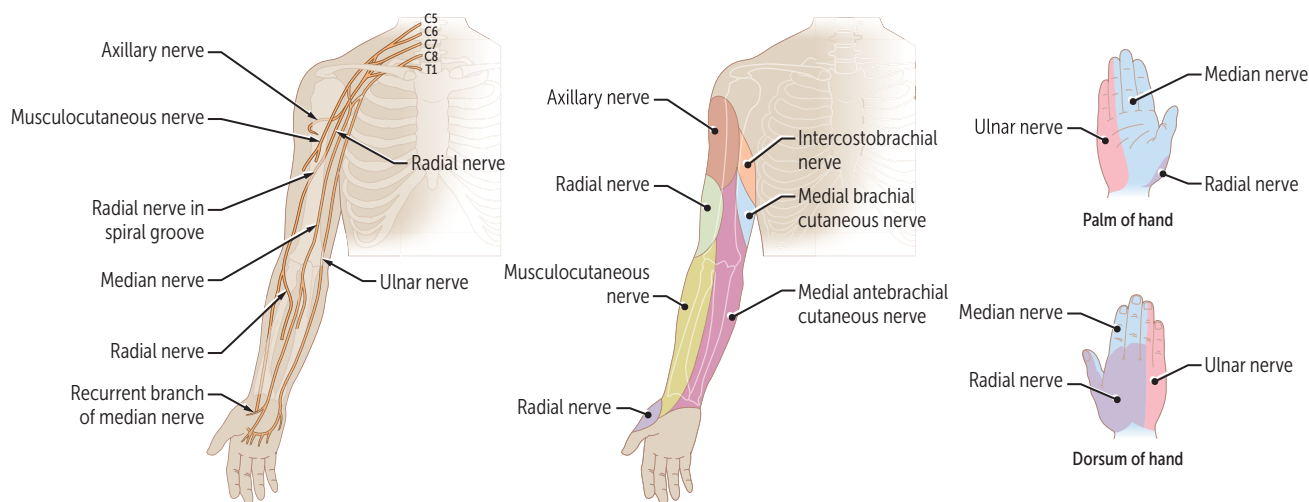
► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—ANATOMY AND PHYSIOLOGY

Upper extremity nerves

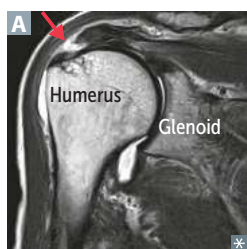
NERVE	CAUSES OF INJURY	PRESENTATION
Axillary (C5-C6)	Fractured surgical neck of humerus Anterior dislocation of humerus	Flattened deltoid Loss of arm abduction at shoulder ($> 15^\circ$) Loss of sensation over deltoid and lateral arm
Musculocutaneous (C5-C7)	Upper trunk compression	↓ biceps (C5-C6) reflex Loss of forearm flexion and supination Loss of sensation over radial and dorsal forearm
Radial (C5-T1)	Compression of axilla, eg, due to crutches or sleeping with arm over chair (“Saturday night palsy”) Midshaft fracture of humerus Repetitive pronation/supination of forearm, eg, due to screwdriver use (“finger drop”)	Injuries above the elbow cause loss of sensation over posterior arm/forearm and dorsal hand, wrist drop (loss of elbow, wrist, and finger extension) with ↓ grip strength (wrist extension necessary for maximal action of flexors) Injuries below the elbow can cause paresthesias of the dorsal hand (superficial radial nerve) or wrist drop (posterior interosseus nerve) Tricep function and posterior arm sensation spared in midshaft fracture
Median (C5-T1)	Supracondylar fracture of humerus → proximal lesion of the nerve Carpal tunnel syndrome and wrist laceration → distal lesion of the nerve	“Ape hand” and “Hand of benediction” Loss of wrist flexion and function of the lateral two L umbricals, O pponens pollicis, A bductor pollicis brevis, F lexor pollicis brevis (LOAF) Loss of sensation over thenar eminence and dorsal and palmar aspects of lateral 3 1/2 fingers with proximal lesion
Ulnar (C8-T1)	Fracture of medial epicondyle of humerus (proximal lesion) Fractured hook of hamate (distal lesion) from fall on outstretched hand Compression of nerve against hamate as the wrist rests on handlebar during cycling	“Ulnar claw” on digit extension Radial deviation of wrist upon flexion (proximal lesion) ↓ flexion of ulnar fingers, abduction and adduction of fingers (interossei), thumb adduction, actions of ulnar 2 lumbrical muscles Loss of sensation over ulnar 1 1/2 fingers including hypothenar eminence
Recurrent branch of median nerve (C5-T1)	Superficial laceration of palm	“Ape hand” Loss of thenar muscle group: opposition, abduction, and flexion of thumb No loss of sensation

Humerus fractures, proximally to distally, follow the **ARM** (Axillary → Radial → Median) nerves

Upper extremity nerves (continued)



Rotator cuff muscles

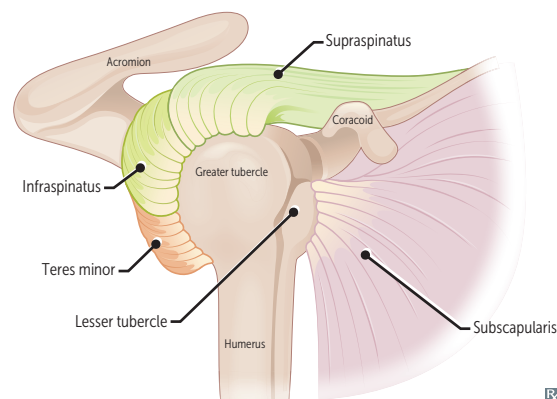


Shoulder muscles that form the rotator cuff:

- **S**upraspinatus (suprascapular nerve)—abducts arm initially (before the action of the deltoid); most common rotator cuff injury (trauma or degeneration and impingement → tendinopathy or tear [arrow in **A**]), assessed by “empty/full can” test
- **I**nfraspinatus (suprascapular nerve)—externally rotates arm; pitching injury
- **t**eres minor (axillary nerve)—adducts and externally rotates arm
- **S**ubscapularis (upper and lower subscapular nerves)—internally rotates and adducts arm

Innervated primarily by C5-C6.

SItS (small t is for teres **minor**).

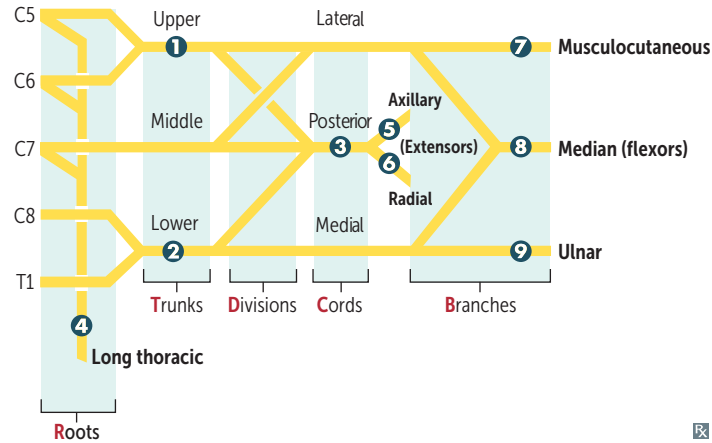


Arm abduction

DEGREE	MUSCLE	NERVE
0°–15°	Supraspinatus	Suprascapular
15°–90°	Deltoid	Axillary
> 90°	Trapezius	Accessory
> 90°	S erratus A nterior	L ong T horacic (SALT)

Brachial plexus lesions

- 1 Erb palsy ("waiter's tip")
- 2 Klumpke palsy (claw hand)
- 3 Wrist drop
- 4 Winged scapula
- 5 Deltoid paralysis
- 6 "Saturday night palsy" (wrist drop)
- 7 Difficulty flexing elbow, variable sensory loss
- 8 Decreased thumb function, "hand of benediction"
- 9 Intrinsic muscles of hand, claw hand



Divisions of brachial plexus:

Remember

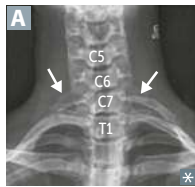
To

Drink

Cold

Beer

Trunks of brachial plexus and the subclavian artery pass between anterior and middle scalene muscles. Subclavian vein passes anteromedial to the scalene triangle.

CONDITION	INJURY	CAUSES	MUSCLE DEFICIT	FUNCTIONAL DEFICIT	PRESENTATION
Erb palsy ("waiter's tip")	Traction or tear of upper trunk : C5-C6 roots	Infants—lateral traction on neck during delivery Adults—trauma leading to neck traction (eg, falling on head and shoulder in motorcycle accident)	D eltoid, supraspinatus I nfra s pinatus, supraspinatus B iceps brachii H erb gets DIBs on t ips	Abduction (arm hangs by side) Lateral rotation (arm medially rotated) Flexion, supination (arm extended and pronated)	
Klumpke palsy	Traction or tear of lower trunk : C8-T1 roots	Infants—upward force on arm during delivery Adults—trauma (eg, grabbing a tree branch to break a fall)	Intrinsic hand muscles: lumbricals, interossei, thenar, hypothenar	C law hand ("Clawmpke" palsy): lumbricals normally flex MCP joints and extend DIP and PIP joints	
Thoracic outlet syndrome	Compression of lower trunk and subclavian vessels, most commonly within the scalene triangle	Cervical/anomalous first ribs (arrows in A), Pancoast tumor	Same as Klumpke palsy	Atrophy of intrinsic hand muscles; ischemia, pain, and edema due to vascular compression	
Winged scapula	Lesion of long thoracic nerve, roots C5-C7 ("wings of heaven")	Axillary node dissection after mastectomy, stab wounds	Serratus anterior	Inability to anchor scapula to thoracic cage → cannot abduct arm above horizontal position B	

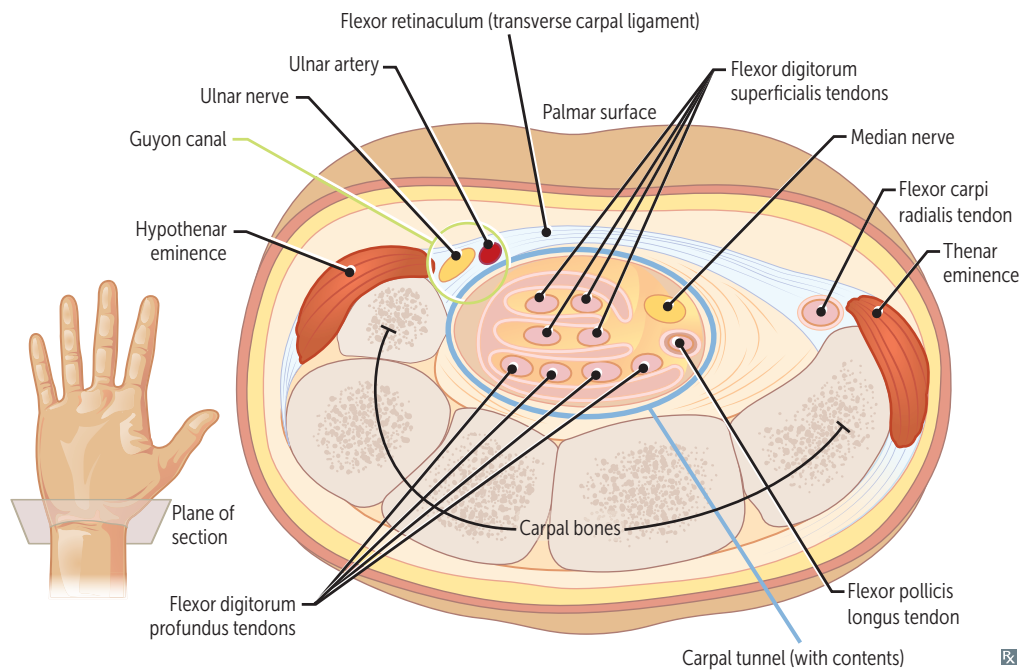
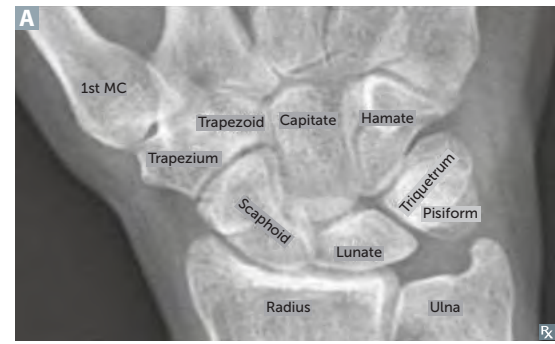
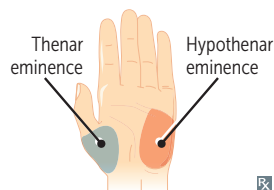
Wrist region

Scaphoid, lunate, triquetrum, pisiform, hamate, capitate, trapezoid, trapezium **A**. (So long to pinky, here comes the thumb)

Scaphoid (palpable in anatomic snuff box **B**) is the most commonly fractured carpal bone, typically due to a fall on an outstretched hand. Complications of proximal scaphoid fractures include avascular necrosis and nonunion due to retrograde blood supply from a branch of the radial artery. Occult fracture not always seen on initial x-ray.

Dislocation of lunate may impinge median nerve and cause carpal tunnel syndrome.

Fracture of the hook of the hamate can cause ulnar nerve compression—**Guyon canal syndrome**.

**Hand muscles**

Thenar (median)—**O**pponens pollicis, **A**bductor pollicis brevis, **F**lexor pollicis brevis: superficial and deep (by ulnar nerve) heads, adductor pollicis (by ulnar nerve).

Hypothenar (ulnar)—**O**pponens digiti minimi, **A**bductor digiti minimi, **F**lexor digiti minimi brevis.

Dorsal interossei (ulnar)—abduct the fingers.

Palmar interossei (ulnar)—adduct the fingers.

Lumbricals (1st/2nd, median; 3rd/4th, ulnar)—flex at the MCP joint, extend PIP and DIP joints.

Both groups perform the same functions: **O**ppose, **A**bduct, and **F**lex (**OAF**).

DAB = **D**orsals **AB**duct.

PAD = **P**almars **AD**duct.

Distortions of the hand

At rest, a balance exists between the extrinsic flexors and extensors of the hand, as well as the intrinsic muscles of the hand—particularly the lumbrical muscles (flexion of MCP, extension of DIP and PIP joints).

“Clawing” **A**—seen best with **distal** lesions of median or ulnar nerves. Remaining extrinsic flexors of the digits exaggerate the loss of the lumbricals → fingers extend at MCP, flex at DIP and PIP joints.

Deficits less pronounced in **proximal** lesions; deficits present during voluntary flexion of the digits.

SIGN	“Ulnar claw”	“Hand of benediction”	“Median claw”	“Trouble making a fist”
PRESENTATION				
CONTEXT	Extending fingers/at rest	Making a fist	Extending fingers/at rest	Making a fist
LOCATION OF LESION	Distal ulnar nerve	Proximal median nerve	Distal median nerve	Proximal ulnar nerve

Note: Atrophy of the thenar eminence can be seen in median nerve lesions, while atrophy of the hypothenar eminence can be seen in ulnar nerve lesions.

Knee exam

Lateral femoral condyle to anterior tibia: **ACL**. **M**edial femoral condyle to posterior tibia: **PCL**. **LAMP**.

TEST**PROCEDURE****Anterior drawer sign**

Positive in ACL tear. Tibia glides anteriorly (relative to femur) when knee is at 90° angle. Alternatively, Lachman test can be done by placing the knee at a 30° angle.

Posterior drawer sign

Bending knee at 90° angle, ↑ posterior gliding of tibia due to PCL injury.

Valgus stress test

Abnormal passive abduction. Knee extended or at ~ 30° angle, lateral (valgus) force → medial space widening of tibia → MCL injury.

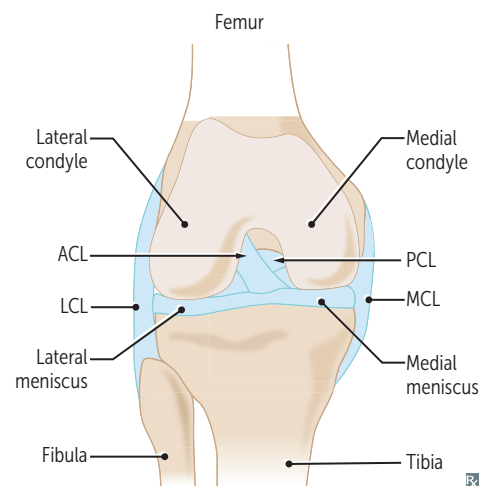
Varus stress test

Abnormal passive adduction. Knee extended or at ~ 30° angle, medial (varus) force → lateral space widening of tibia → LCL injury.

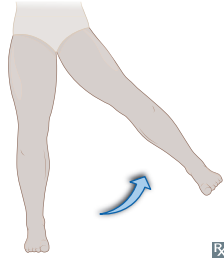
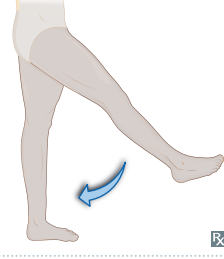
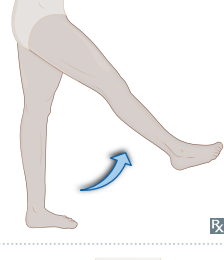
McMurray test

During flexion and extension of knee with rotation of tibia/foot (**LIME**):

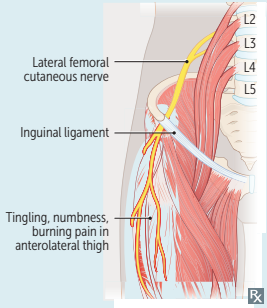
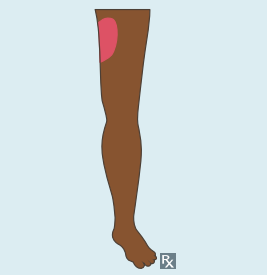
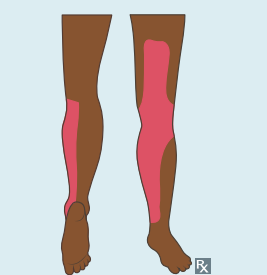
- Pain, “popping” on internal rotation and varus force → **L**ateral meniscal tear (**I**nternal rotation stresses lateral meniscus)
- Pain, “popping” on external rotation and valgus force → **M**edial meniscal tear (**E**xternal rotation stresses medial meniscus)



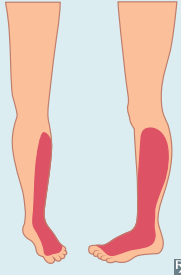
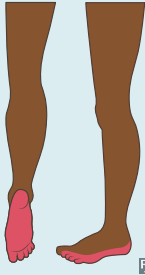
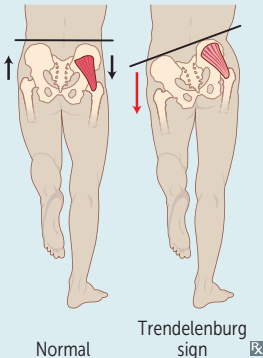
Actions of hip muscles

ACTION	MUSCLES	PRESENTATION
Abductors	Gluteus medius, gluteus minimus	
Adductors	Adductor magnus, adductor longus, adductor brevis	
Extensors	Gluteus maximus, semitendinosus, semimembranosus, long head of biceps femoris	
Flexors	Iliopsoas (iliacus and psoas), rectus femoris, tensor fascia lata, pectineus, sartorius	
Internal rotation	Gluteus medius, gluteus minimus, tensor fascia latae	
External rotation	Iliopsoas, gluteus maximus, piriformis, obturator internus, obturator externus	

Lower extremity nerves

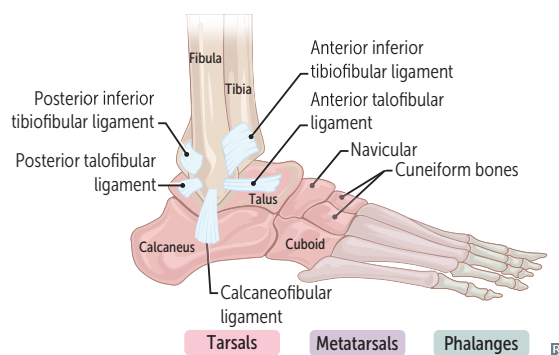
NERVE	INNERVATION	CAUSE OF INJURY	PRESENTATION/COMMENTS
Iliohypogastric (T12-L1)	Sensory—suprapubic region Motor—transversus abdominis and internal oblique	Abdominal surgery (commonly inguinal hernia repair)	Neuropathic pain (burning or tingling) in surgical incision site radiating to inguinal and suprapubic region
Genitofemoral nerve (L1-L2)	Sensory—scrotum/labia majora, medial thigh Motor—cremaster	Laparoscopic surgery	↓ upper medial thigh and anterior thigh sensation beneath the inguinal ligament (lateral part of the femoral triangle); absent cremasteric reflex
Lateral femoral cutaneous (L2-L3) 	Sensory—anterior and lateral thigh	Tight clothing, obesity, pregnancy, pelvic procedures	↓ thigh sensation (anterior and lateral) Meralgia paresthetica —compression of lateral femoral cutaneous nerve → tingling, numbness, burning pain in anterolateral thigh
Obturator (L2-L4) 	Sensory—medial thigh Motor—obturator externus, adductor longus, adductor brevis, gracilis, pectineus, adductor magnus	Pelvic operation	↓ thigh sensation (medial) and adduction
Femoral (L2-L4) 	Sensory—anterior thigh, medial leg Motor—quadriceps, iliacus, pectineus, sartorius	Pelvic fracture, compression from retroperitoneal hematoma or psoas abscess	↓ leg extension (↓ patellar reflex)
Sciatic (L4-S3)	Sensory—posterior thigh, posterior knee, and all below knee (except narrow band on medial lower leg) Motor—semitendinosus, semimembranosus, biceps femoris, adductor magnus	Herniated disc, posterior hip dislocation, piriformis syndrome	Splits into common peroneal and tibial nerves

Lower extremity nerves (*continued*)

NERVE	INNERVATION	CAUSE OF INJURY	PRESENTATION/COMMENTS
Common (fibular) peroneal (L4-S2) 	Superficial peroneal nerve: ▪ Sensory—dorsum of foot (except webspace between hallux and 2nd digit) ▪ Motor—peroneus longus and brevis Deep peroneal nerve: ▪ Sensory—webspace between hallux and 2nd digit ▪ Motor—tibialis anterior	Trauma or compression of lateral aspect of leg, fibular neck fracture	PED = Peroneal Everts and Dorsiflexes; if injured, foot drop Loss of sensation on dorsum of foot Foot drop —inverted and plantarflexed at rest, loss of eversion and dorsiflexion; “steppage gait”
Tibial (L4-S3) 	Sensory—sole of foot Motor—biceps femoris (long head), triceps surae, plantaris, popliteus, flexor muscles of foot	Knee trauma, Baker cyst (proximal lesion); tarsal tunnel syndrome (distal lesion)	TIP = Tibial Inverts and Plantarflexes; if injured, can't stand on TIP toes Inability to curl toes and loss of sensation on sole; in proximal lesions, foot everted at rest with weakened inversion and plantar flexion
Superior gluteal (L4-S1) 	Motor—gluteus medius, gluteus minimus, tensor fascia latae	Iatrogenic injury during intramuscular injection to superomedial gluteal region (prevent by choosing superolateral quadrant, preferably anterolateral region)	Trendelenburg sign/gait—pelvis tilts because weight-bearing leg cannot maintain alignment of pelvis through hip abduction Lesion located on side of raised hip, ipsilateral to standing leg
Inferior gluteal (L5-S2)	Motor—gluteus maximus	Posterior hip dislocation	Difficulty climbing stairs, rising from seated position; loss of hip extension
Pudendal (S2-S4)	Sensory—perineum Motor—external urethral and anal sphincters	Stretch injury during childbirth, prolonged cycling, horseback riding	↓ sensation in perineum and genital area; can cause fecal and/or urinary incontinence Can be blocked with local anesthetic during childbirth using ischial spine as a landmark for injection

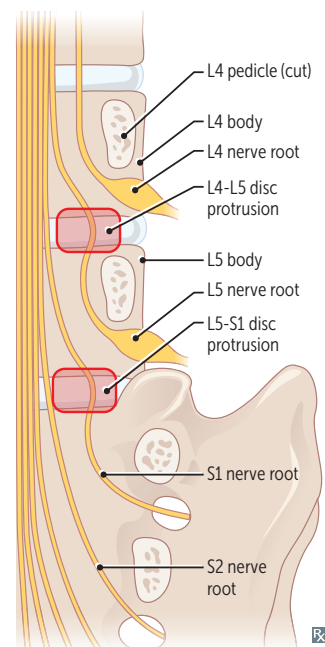
Ankle sprains

Anterior talofibular ligament—most common ankle sprain overall, classified as a **low** ankle sprain. Due to overinversion/supination of foot. Anterior inferior tibiofibular ligament—most common **high** ankle sprain. **High tide**.

**Signs of lumbosacral radiculopathy**

Paresthesia and weakness related to specific lumbosacral spinal nerves. Intervertebral disc (nucleus pulposus) herniates posterolaterally through annulus fibrosus (outer ring) into spinal canal due to thin posterior longitudinal ligament and thicker anterior longitudinal ligament along midline of vertebral bodies. Nerve affected is usually below the level of herniation. ⊕ straight leg raise, ⊕ contralateral straight leg raise, ⊕ reverse straight leg raise (femoral stretch).

Disc level herniation	L3-L4	L4-L5	L5-S1
Nerve root affected	L4	L5	S1
Dermatome affected			
Clinical findings	Weakness of knee extension ↓ patellar reflex	Weakness of dorsiflexion Difficulty in heel walking	Weakness of plantar flexion Difficulty in toe walking ↓ Achilles reflex

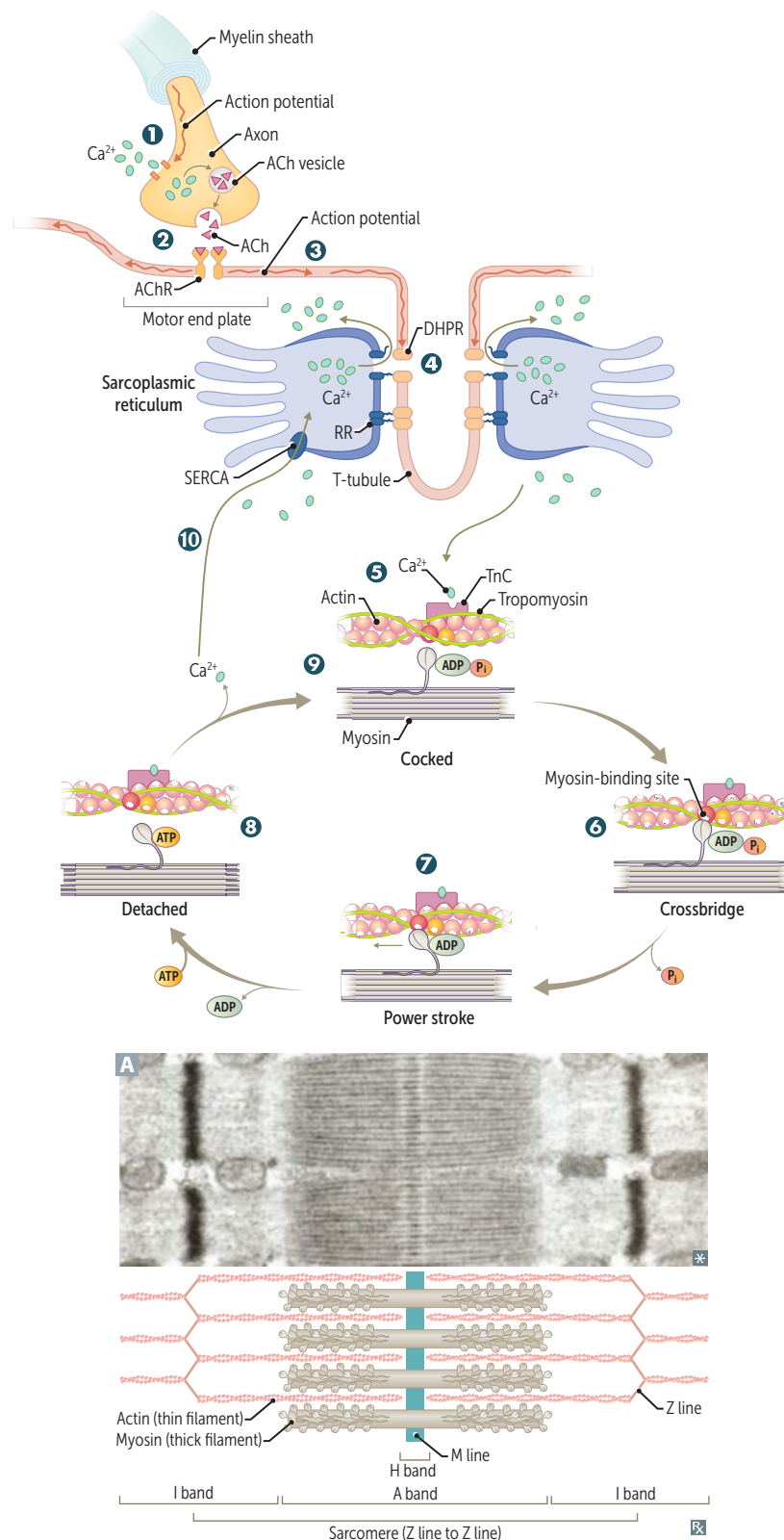
**Neurovascular pairing**

Nerves and arteries are frequently named together by the bones/regions with which they are associated. The following are exceptions to this naming convention.

LOCATION	NERVE	ARTERY
Axilla/lateral thorax	Long thoracic	Lateral thoracic
Surgical neck of humerus	Axillary	Posterior circumflex
Midshaft of humerus	Radial	Deep brachial
Distal humerus/cubital fossa	Median	Brachial
Popliteal fossa	Tibial	Popliteal
Posterior to medial malleolus	Tibial	Posterior tibial

Motor neuron action potential to muscle contraction

T-tubules are extensions of plasma membrane in contact with the sarcoplasmic reticulum, allowing for coordinated contraction of striated muscles.



- 1 Action potential opens presynaptic voltage-gated Ca^{2+} channels, inducing acetylcholine (ACh) release.
- 2 Postsynaptic ACh binding leads to muscle cell depolarization at the motor end plate.
- 3 Depolarization travels over the entire muscle cell and deep into the muscle via the T-tubules.
- 4 Membrane depolarization induces conformational changes in the voltage-sensitive dihydropyridine receptor (DHPR) and its mechanically coupled ryanodine receptor (RR) → Ca^{2+} release from the sarcoplasmic reticulum (buffered by calsequestrin) into the cytoplasm.
- 5 Tropomyosin is blocking myosin-binding sites on the actin filament. Released Ca^{2+} binds to troponin C (TnC), shifting tropomyosin to expose the myosin-binding sites.
- 6 Myosin head binds strongly to actin (crossbridge). P_i released, initiating power stroke.
- 7 During the power stroke, force is produced as myosin pulls on the thin filament **A**. Muscle shortening occurs, with shortening of **H** and **I** bands and between **Z** lines (**HI**, I'm wearing short**Z**). The **A** band remains the same length (**A** band is **A**lways the same length). ADP is released at the end of the power stroke.
- 8 Binding of new ATP molecule causes detachment of myosin head from actin filament. Ca^{2+} is resealed.
- 9 ATP hydrolysis into ADP and P_i results in myosin head returning to high-energy position (cocked). The myosin head can bind to a new site on actin to form a crossbridge if Ca^{2+} remains available.
- 10 Reuptake of calcium by sarco(endo)plasmic reticulum Ca^{2+} ATPase (SERCA) → muscle relaxation.

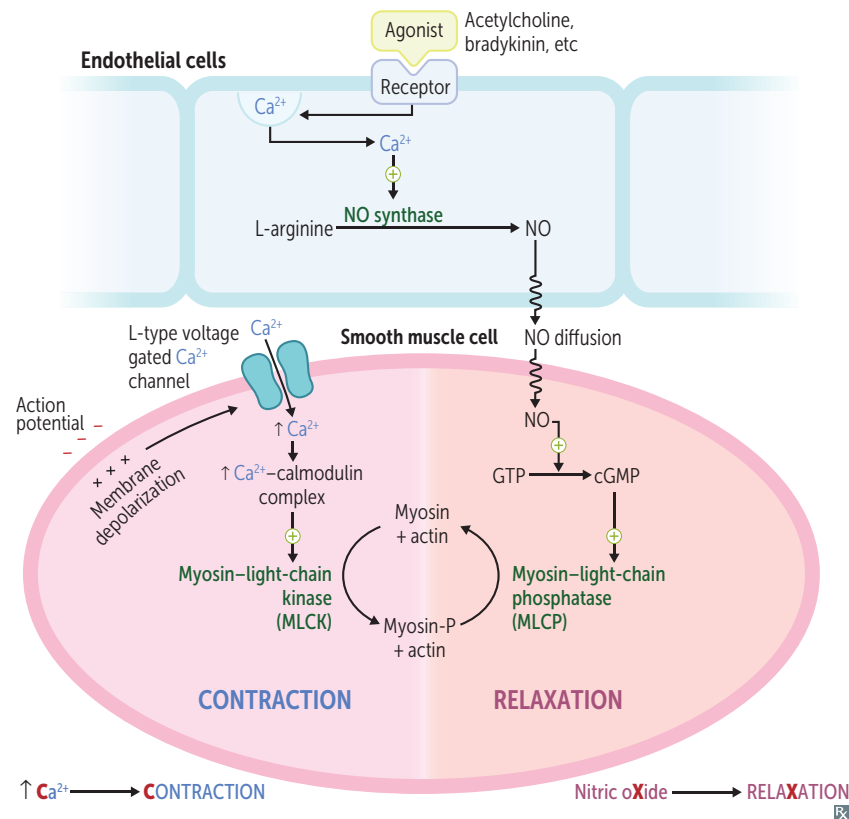
Types of skeletal muscle fibers

Two types, normally distributed randomly within muscle. Muscle fiber type grouping commonly occurs due to reinnervation of denervated muscle fibers in peripheral nerve damage.

	Type I	Type II
CONTRACTION VELOCITY	Slow	Fast
FIBER COLOR	Red	White
PREDOMINANT METABOLISM	Oxidative phosphorylation → sustained contraction	Anaerobic glycolysis
MITOCHONDRIA, MYOGLOBIN	↑	↓
TYPE OF TRAINING	Endurance training	Weight/resistance training, sprinting
NOTES	Think “1 slow red ox”	Think “2 fast white antelopes”

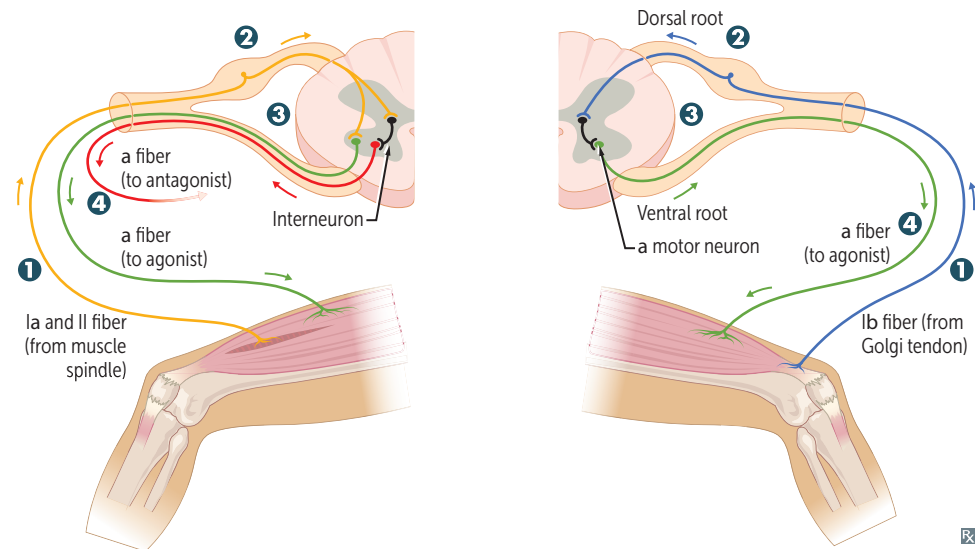
Skeletal muscle adaptations

	Atrophy	Hypertrophy
MYOFIBRILS	↓ (removal via ubiquitin-proteasome system)	↑ (addition of sarcomeres in parallel)
MYONUCLEI	↓ (selective apoptosis)	↑ (fusion of satellite cells, which repair damaged myofibrils; absent in cardiac muscles)

Vascular smooth muscle contraction and relaxation

Muscle proprioceptors Specialized sensory receptors that relay information about muscle dynamics.

	Muscle stretch receptors	Golgi tendon organ
PATHWAY	➊ ↑ length and speed of stretch → ➋ via dorsal root ganglion (DRG) → ➌ activation of inhibitory interneuron and α motor neuron → ➍ simultaneous inhibition of antagonist muscle (prevents overstretching) and activation of agonist muscle (contraction).	➊ ↑ tension → ➋ via DRG → ➌ activation of inhibitory interneuron → ➍ inhibition of agonist muscle (reduced tension within muscle and tendon)
LOCATION/INNERVATION	Body of muscle/type Ia and II sensory axons	Tendons/type Ib sensory axons
ACTIVATION BY	↑ muscle stretch. Responsible for deep tendon reflexes	↑ muscle tension

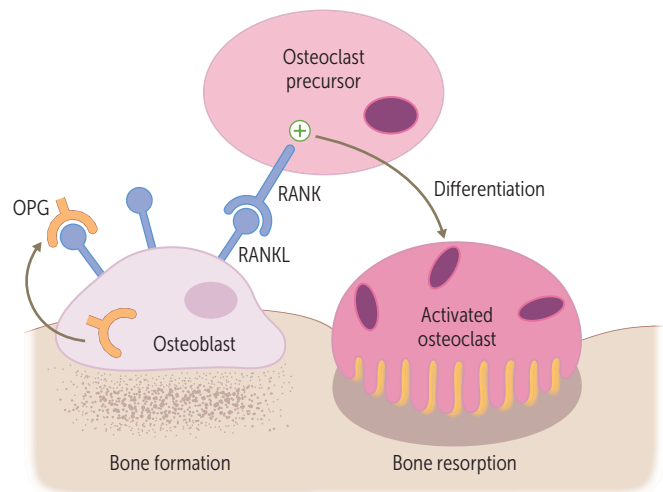


Bone formation

Endochondral ossification	Bones of axial skeleton, appendicular skeleton, and base of skull. Cartilaginous model of bone is first made by chondrocytes. Osteoclasts and osteoblasts later replace with woven bone and then remodel to lamellar bone. In adults, woven bone occurs after fractures and in Paget disease. Defective in achondroplasia.
Membranous ossification	Bones of calvarium, facial bones, and clavicle. Woven bone formed directly without cartilage. Later remodeled to lamellar bone.

Cell biology of bone

Osteoblast	B uilds b one by secreting collagen and catalyzing mineralization in alkaline environment via ALP. Differentiates from mesenchymal stem cells in periosteum. Osteoblastic activity measured by bone ALP, osteocalcin, propeptides of type I procollagen.
Osteoclast	Dissolves (“ c rushes”) bone by secreting H^+ and collagenases. Differentiates from a fusion of monocyte/macrophage lineage precursors. RANK receptors on osteoclasts are stimulated by RANKL (RANK ligand, expressed on osteoblasts). OPG (osteoprotegerin, a RANKL decoy receptor) binds RANKL to prevent RANK-RANKL interaction → ↓ osteoclast activity.
Parathyroid hormone	At low, intermittent levels, exerts anabolic effects (building bone) on osteoblasts and osteoclasts (indirect). Chronically ↑ PTH levels (1° hyperparathyroidism) cause catabolic effects (osteitis fibrosa cystica).
Estrogen	Inhibits apoptosis in bone-forming osteoblasts and induces apoptosis in bone-resorbing osteoclasts. Causes closure of epiphyseal plate during puberty. Estrogen deficiency (surgical or postmenopausal) → ↑ cycles of remodeling and bone resorption → ↑ risk of osteoporosis.

**► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—PATHOLOGY****Overuse injuries of the elbow**

Medial (golfer's) elbow tendinopathy	Repetitive wrist flexion or idiopathic → pain near medial epicondyle.
Lateral (tennis) elbow tendinopathy	Repetitive wrist extension (backhand shots) or idiopathic → pain near lateral epicondyle.